

Complete audit, recovery of trapped source files, reorganization, and verification

Critical Finding: Missing Working Tree Code

Upon initial clone, the entire application was inoperable because the source tree was committed as a compressed zip archive (`src.zip`) rather than tracked files. Both `index.html` (entry point for Vite) and `server.ts` (entry point for Express) referenced non-existent paths. Additionally, conflicting prototype files and peer dependency resolution locks prevented clean local execution.

1. Full Audit of Missing & Modified Items

File / Path	Action	Description & Reason
<code>src/App.tsx</code>	RESTORED	Extracted from <code>src.zip</code> . Full 1,818-line React SPA with authentication, dashboard, reporting, and priority breakdown.
<code>src/main.tsx</code>	RESTORED	Extracted from <code>src.zip</code> . Client entry point required by <code>index.html</code> to bootstrap React DOM.
<code>src/index.css</code>	RESTORED	Extracted from <code>src.zip</code> . Core CSS variables, Tailwind configuration, and scanner animations.
<code>src/backend/routes.ts</code>	RESTORED	Extracted from <code>src.zip</code> . 829-line REST API endpoints (Auth, Issues, GIS reports, Dashboard metrics).
<code>src/backend/ai.ts</code>	RESTORED	Extracted from <code>src.zip</code> . YOLOv8 visual classifier and Gemini Flash vision integration.
<code>src/backend/auth.ts</code>	RESTORED	Extracted from <code>src.zip</code> . JWT authentication and salted bcrypt password hashing security.
<code>src/backend/db.ts</code>	RESTORED	Extracted from <code>src.zip</code> . Atomic persistent JSON database engine modeling 11 municipal collections.
<code>src/backend/duplicate.ts</code>	RESTORED	Extracted from <code>src.zip</code> . GIS Haversine spatial duplicate detector with confidence scoring.
<code>src/backend/priority.ts</code>	RESTORED	Extracted from <code>src.zip</code> . Weighted 5-factor priority calculation engine (0–100 scale).
<code>src/backend/types.ts</code>	RESTORED	Extracted from <code>src.zip</code> . Shared TypeScript models and schema definitions.
<code>prototype/</code>	ADDED	Dedicated directory organizing standalone vanilla web prototype (<code>idea3.html</code> , <code>idea3.css</code> , <code>idea3.js</code>).
<code>server.ts</code>	MODIFIED	Added static route <code>app.use('/prototype', ...)</code> to serve the vanilla frontend concurrently with the React SPA.

File / Path	Action	Description & Reason
package.json	MODIFIED	Updated package name to civic-eye , version 1.0.0 , removed conflicting esbuild peer dependency, and organized devDependencies.
vite.config.ts	MODIFIED	Migrated deprecated CJS __dirname to native ESM import.meta.dirname .
.gitignore	MODIFIED	Added data/ local JSON database folder to prevent committing transient test data.
README.md	MODIFIED	Updated architecture diagram and quickstart references to reflect prototype/idea3.html .
src.zip	REMOVED	Removed redundant 36 KB binary archive after extracting all source files into the working tree.
idea 3.* (with spaces)	REMOVED	Deleted unintegrated mock prototype files containing spaces in filenames (idea 3.css , idea 3.html , idea 3.js).

2. Cleaned Repository Structure

The project is now cleanly segregated between root build configs, standalone prototypes, and full-stack source code:

```
civic-eye/
├── index.html           # Vite entry point for React SPA
├── package.json         # Dependency definitions & scripts (civic-eye v1.0.0)
├── tsconfig.json        # TypeScript compiler configuration
├── vite.config.ts       # Vite build & bundler configuration (ESM dirname)
├── server.ts            # Express API server + Vite dev middleware
├── metadata.json        # Platform capability metadata
├── README.md            # Project architecture & API specifications
├── .env.example         # Environment variable templates
├── .gitignore           # Ignores node_modules, dist, .env, and data/
├── prototype/           # Dedicated Standalone Vanilla Prototype
│   ├── idea3.html       # Semantic HTML civic reporting interface
│   ├── idea3.css        # Custom UI styling & responsive layout
│   └── idea3.js          # Vanilla JS wired to /api REST endpoints
└── src/                 # Core Source Code (Restored & Verified)
    ├── App.tsx          # Main React SPA component
    ├── main.tsx         # React 19 DOM bootstrap
    ├── index.css        # Global CSS & Tailwind utilities
    └── backend/         # Node.js + Express backend services
        ├── ai.ts        # YOLOv8 / Gemini Flash computer vision
        ├── auth.ts      # JWT + Salted bcrypt authentication & RBAC
        ├── db.ts        # Atomic disk persistence engine (data/civiceye_db.json)
        ├── duplicate.ts  # GIS Haversine spatial duplicate clustering
        ├── priority.ts   # Multi-factor priority score algorithm
        ├── routes.ts     # Express REST route controllers (/api)
        └── types.ts      # Domain interfaces & database types
```

3. Dual Frontend Architecture

1. Primary React SPA

Modern React 19 + Tailwind SPA mounted at root.

- **URL:** `http://localhost:3000/`
- **Features:** Interactive role switcher, real-time statistics filters, AI scanner animation, transparent priority formula breakdown dialog.
- **Source:** `src/App.tsx`

2. Standalone Vanilla Prototype

Self-contained HTML5/CSS3/ES6 prototype.

- **URL:** `http://localhost:3000/prototype/idea3.html`
- **Features:** Lightweight, zero-build-step client, custom in-page toast notifications, directly callable offline.
- **Source:** `prototype/idea3.html`

4. Verification & Validation Results

All Verification Checks Passed Cleanly

Check	Command	Result	Notes
Dependency Installation	<code>npm install</code>	Passed	194 packages installed with resolved peer dependencies.
TypeScript Type Check	<code>npm run lint</code> (<code>tsc --noEmit</code>)	Passed	0 errors across entire frontend and backend TypeScript code.
Production Build	<code>npm run build</code> (vite build)	Passed	Compiled client bundle in 83ms with 0 warnings.
IDE Integration	<code>open -a "Antigravity IDE"</code>	Passed	Workspace opened at <code>/Users/manish/.gemini/antigravity/scratch/civic-eye</code> .